FULL VOLUME

Plain-Jane speakers can no longer match today's demanding audio applications. You deserve better!

A good set of speakers not only sounds good, but feels good too. There is nothing that can stir the senses more than a delicately rendered passage of Mozart or, if you are inclined towards gaming, a gunfight where you almost feel the recoil of the shotgun with bullets ricocheting all around you!

No longer are home computers sold just as simple devices for getting your tax work done and managing your weekly expenditures—they are meant to be full-blown entertainment centres. You can use a PC as a massive repository for all your family's music or double it up as a home theatre system for movies on the weekend. Today when you buy a PC, you can easily integrate a home theatre system with it, and that too at a budget.

In this comparison test, we have deliberately not tested any standard stereo speakers. We have only tested speakers that bundle a subwoofer with them. This, we feel, is a critical component in delivering the power and the room-filling sound that is a requisite for today's demanding audio applications and an immersive home theatre experience. The fact that even the most basic 2.1 speaker systems are available for under Rs 2,000 is testament enough that the standard of the audio component for the PC is on the rise, even at the most basic level.

Digit Test Process

We used a Pentium 4 2.4 GHz, on an Intel D850EMV2 motherboard with 256 MB of RDRAM running Windows XP Professional as the test system for evaluating the speakers. Since sound quality depends as much on the soundcard as the output speakers, we used the Sound Blaster Audigy Platinum soundcard for all our tests as it provides a pure audio signal with very little noise. We tested speakers across three categories:

2.1 speaker systems: These speakers feature a front left and front right channel along with the subwoofer channel used for low frequency sound output. At their basic level, however, they are still stereo speakers as they have only two channels of input (the subwoofer channel is derived from the stereo channel).

4.1 speaker systems: These speakers can output four discrete channels of sound—the front left, front right, rear left and rear right speakers along with the subwoofer channel used for producing bass frequencies. These speakers have four channels of input and the bass channel is derived from the front channels.

5.1 speaker systems: These speakers have six discrete channels of audio—the front left, front right, centre, rear left and rear right speakers along with a dedicated subwoofer channel used for handling bass frequencies. They can accept six separate analogue or a digital input channel.

To obtain the purest possible sound without any form of enhancement, any available equalisation on the speakers was turned off and the volume, bass and treble levels were kept at a fixed 50 per cent throughout, except in the power-handling test, where the volume was raised to its maximum possible setting for evaluating the sound quality at high output levels. The test was run in an isolated room to avoid any external audio disturbance.

Test Methodology

We evaluated the speakers on the basis of features (20 per cent weightage), performance (50 per cent weightage), value for money (20 per cent weightage) and the warranty and support offered (10 per cent weightage).

Features: We awarded points based on the capability and the features of the speakers. We logged technical specifications such as the power rating, frequency response range, the number and type of connectors provided such as S/PDIF, coaxial, optical and any other type of analogue or digital inputs. We also checked whether the speakers adhered to any standards such as Dolby Digital or THX and they were given extra points if they did. Additional accessories such as speaker stands, remote controllers and even longer bundled speaker cables were awarded separate points.

Performance: This was the single most important category as we tested the ability of speaker sets to deliver on their claimed audio performance. Performance was evaluated on the basis of the purity of the sound reproduced by the speakers under various conditions using an array of tests.

a) Sound reproduction quality: We evaluated the performance of the speakers under various applications such as gaming, music and movie playback. The in-game audio tests checked the overall clarity of the speakers while reproducing in-game music along with the other gaming sound effects. We used the retail version of *Quake III Arena* v1.30 and *Serious Sam: Second Encounter* for this test. We then tested the ability of the speakers to effectively reproduce surround sound in movie playback. For testing overall surround sound quality, we played a VOB file. This is a file that is encoded with DVD-quality video (usually at a resolution of 720x480) with full support for Dolby Digital audio. Next, we tested the music playback quality using two types of audio files—CD audio and MP3. We used two tracks to evaluate this: 'Freak on a Leash' by Korn and the Hindi track 'Mitwa' from the movie *Lagaan*. Finally, we used a special test Audio CD that contained reference audio files consisting of a cross-section of music and sound effect files. This CD was used to ascertain the response of the speakers to different types of audio and to transient sound effects such as gunshots and explosions.

b) Sound frequency test: This test was divided into four parts in which we played four different test tones. Each of these files were in frequencies ranging between a low 30 Hz to a maximum of 15 KHz. Points were given based on the speaker's ability to accurately reproduce the sound in each frequency range.